

- 12 (a) (i)** electron diffraction/electron microscope (allow other sensible suggestions) B1 [1]
- (ii) photoelectric effect/Compton scattering (allow other sensible suggestions) B1 [1]
- (b) (i)** arrow clear from -0.54 eV to -3.40 eV B1 [1]
- (ii) $E = hc/\lambda$ or $E = hf$ and $c = f\lambda$ C1
- $\lambda = (6.63 \times 10^{-34} \times 3.00 \times 10^8) / [(3.40 - 0.54) \times 1.60 \times 10^{-19}] = 4.35 \times 10^{-7} \text{ m}$ A1 [2]
- (c) (i)** wavelength associated with a particle M1
that is moving/has momentum/has speed/has velocity A1 [2]
- (ii) $\lambda = h/mv$
- $v = (6.63 \times 10^{-34}) / (9.11 \times 10^{-31} \times 4.35 \times 10^{-7})$ C1
- $= 1.67 \times 10^3 \text{ ms}^{-1}$ A1 [2]